25900198 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCSTA100 - INTRODUCTION TO STATISTICAL MODELLING

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **7** questions

Each question carries **2** marks

1. Given the pmf as f(x) = k x3; x = 1, 2, 3 and 0 elsewhere, find k. [K] / [CO1]
2. Mention any two properties of Moment Generating Function. [U] / [CO2]
3. In a binomial distribution, the mean is 5 and the variance is 0.3. Compute the value of p. [K] / [CO3]
4. Given the pdf as f(x) = 6 x (1 - x) ; 0 < x < 1 and 0 elsewhere, find pdf of Y = 3X . [U] / [CO1]
5. The probability that sets consisting of 1, 2, 3, 4, and 5 persons pay a visit to an art gallery are 0.2, 0.5, 0.2, 0.07 and 0.03 respectively. What is the expected number of persons per set? [An] / [CO2]
6. A basketball player can throw a ball into the basket with probability 0.6. What is the probability that he misses the throw? [U] / [CO3]

7. Consider the random experiment of inspecting product packets of a factory in a day. Classify the random variable defined as 'the number of defectives in a packet' as discrete or continuous. Justify your answer. [K] / [CO1]

8. Write down the p.d.f. of Normal distribution . [U] / [CO3]

9. A telephone number is selected at random from a dictionary. Suppose X denote the last digit of selected telephone number. Find the probability that the last digit of the selected number is 6. [A] / [CO3]

10. Write down the p.d.f. of normal distribution with mean 100 and variance 25. [U] / [CO3]

[2x7 = 14]

Part B

Short Essay Questions

Answer any **4** questions

Each question carries **6** marks

11. Consider the following pdfs . Find the value of k in both cases. [K] / [CO1]
    (1) $f(x) = kx^2$; $0 < x < 1$ and 0 elsewhere
    (2) $f(x) = k \times (1 - x)$; $0 < x < 1$ and 0 elsewhere

12. If pdf of X is $f(x)=2x$; $0<x<1$ and 0 elsewhere, find E(X) and V(X). [A] / [CO2]

13. If X has binomial distribution with mean 6 and variance 3.6, find P(X = 2). [A] / [CO3]

14. What are the uses of Poisson distribution? [U] / [CO3]

15. Define exponential distribution. Mention any three properties of it. [A] / [CO3]

16. Define geometric distribution. Mention its properties. [A] / [CO3]

[6x4 = 24]

Part C

Essay Questions

Answer any **1** questions

Each question carries **12** marks

17. Consider the random experiment of throwing an unbiased die and let X be the random variable denoting the number shown by the die. Obtain the pmf of X. Also find the distribution function and sketch it graphically. [K] / [CO1]

18. In a certain factory turning out optical lenses, there is a small chance of 1/500 for any lens to be defective. The lenses are supplied in packets of 40. Use Poisson distribution to calculate the approximate number of packets containing no defective, one defective and three defectives, in a consignment of 20000 packets. [A] / [CO3]

[12x1 = 12]